ABSTRACT

Introduction. Salt has been a pillar of Himalayan and Tibetan life, sustaining economies, shaping rituals, supporting multi-commodity networks and serving as an exchange token among nomads. This paper explores the politics of the Himalayan salt economy, arguing that salt negotiated power and territorial control.

Method. The study uses historical archives, colonial gazetteers and administrative reports from the Himalayas to develop social-science literature around salt. Named Entity Recognition and relational-extraction models build a graph linking salt with entities, commodities, communities and administrative blocks.

Analysis. The identified relations were analyzed as a weighted network through validated, concept-excluded and trade-oriented graph views. Salt's position was examined using centrality, brokerage, robustness and removal metrics.

Results. Salt emerged as a structural connector. Rather than appearing only as a commodity, it connected communities, trade goods, routes, places and institutions. Its position remained stable after validation and trade-oriented filtering, showing that salt bridged subsistence exchange, administrative control and multicommodity circulation.

Conclusion. The political conceptualisation of salt presents circulation not merely as commerce or territorial practice, but as infrastructure of control, contestation, sovereignty, dependency and reordering along the Himalayan frontier. British regulation through monopoly, taxation and surveillance transformed long-standing mobility and exchange systems into imperial circuits of extraction.

INTRODUCTION

The Himalayan frontier, especially Ladakh and Tibet, has long been a zone of political contestation involving the Namgyal Dynasty, the Dogras, the British administration, and the Ganden Phodrang government, along with the Chinese state (Bergmann, 2016; Marshall, 2004). Yet, goods such as salt, pashm, tea, and wool have moved throughout the region with relative ease and persistence (Rizvi, 1996, 1995, 1999). These goods have acted as strategic connectors between local economies and imperial governance enabling both economic sustenance and political negotiation (Alam, 2007). Salt from the Changthang lakes supported pastoral communities such as the Rupshu, *Drokpas*, who exchanged it for grain, barley and other low altitude commodities. Before colonial interventions and salt's entanglement with questions of access, taxation, territorial jurisdiction etc. it moved through reciprocal circuits, and successive bartering was a part of the precise calendar as a function of climatic conditions and perfectly regulated trade routes (Fisher, 1986). The pastoral nomads, *Jada*, *Khompas*, and

Changpas, along with their salt and sheep flocks, travelled up to Baltistan via Chevri-Sakti Valley and Tingmosgang in order to secure grains and apricots (Rizvi, 1999; Shankar, n.d.). The annual *Dazem* fair in Himachal provided markets for tea and sugar.

The three principal salt sources of the Himalayan frontier, i.e., the Changthang lakes harvested by Rupshus and *Changpas;* Amdo Tsaidom exploited by the *Golaks* and Mongols; and the Tsalkho Salt pans managed by *Khompas* and Tibetans (Ahmed, 1999; Lecoq, 2022). These zones were also sites of periodic political and territorial dispute, including conflicts between Kharnak groups backed by Hinis Gompas and Rupshus in Changthang, as well as struggles over Ganden Phodrang dominance in regional markets (Ahmed, 1999; Dollfus, 2013). Salt circulation also intersected political arrangements such as the Treaty of Tingmosgang regulating commercial privileges in Ladakh and Tibet, while nineteenth-century arrangements concerning the Mandi salt mines increased administrative significance of salt.

This paper therefore argues that salt is not a mere tradable object, it is a *keystone commodity* of Himalayan political economy which also makes it an infrastructure of governance (Field, 1959). It analyzes how the salt pathways in the Western Himalayas constituted the structural framework for a multi-product commercial network, thus shedding light on the political economy of Himalayan salt trade in the twentieth century. In order to validate the above hypothesis, the paper employs Named Entity Recognition and relation extraction techniques to develop a graph from historical archives, colonial gazetteers, and official reports.

HISTORICAL COMPUTATIONAL BACKGROUND

Himalayan Salt & Multi Commodity Exchange

Ladakh, located along the Himalayan Karakoram arc, and adjoining the high plateaus of Western Tibet, majorly forms the Himalayan salt frontier (Thomson, 1852). These high-altitude crossroads connecting Tibet, Kashmir, and North India functioned as systems of seasonal pastoral mobility and commodity circulation as well as kinship through *netsang* or fictive trade relations, reciprocity through repeated barter, hospitality, and obligation, and negotiation over access, route rights, taxation, and political authority (Rizvi, 1999). The salt lakes of Tso Kar in the Changthang plateau have sustained both the *Drokpa* herders and the Rupshu nomads who traded salt and pashm in exchange for grains, barley, apricots, and spices from lower Ladakh and Zanskar (Ahmed, 1999).

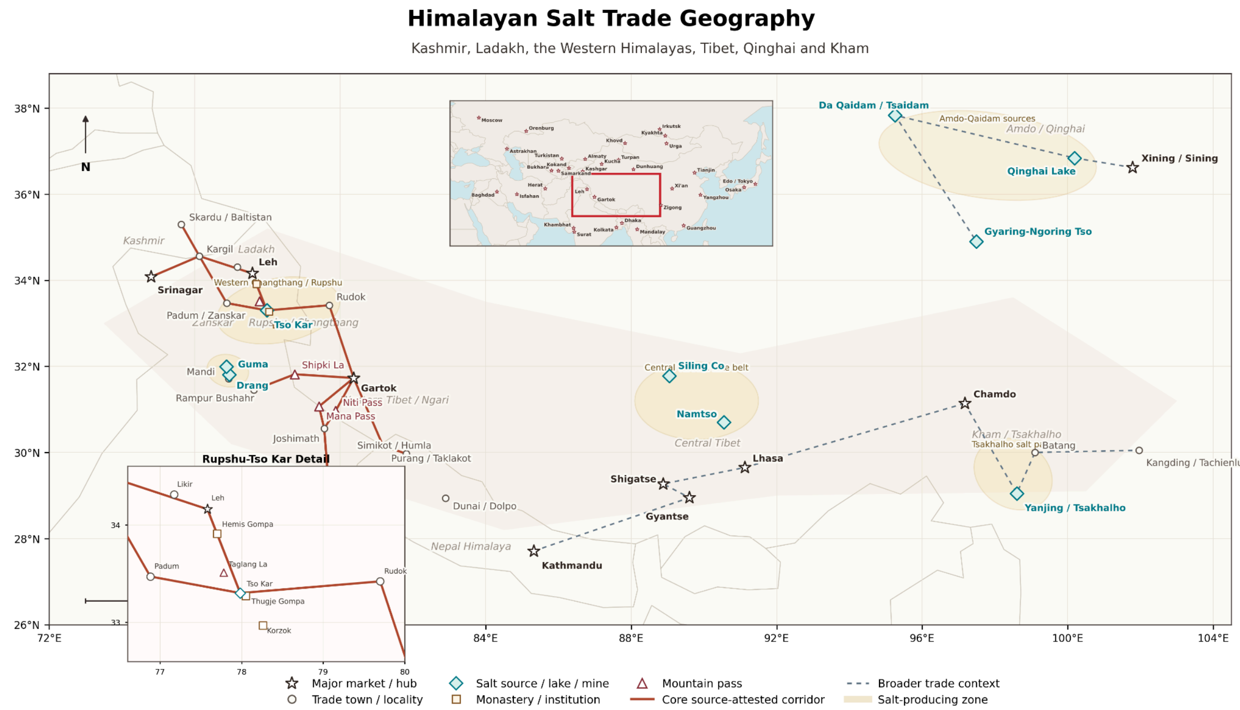

Figure 1. Himalayan salt trade geography across Kashmir, Ladakh, Western Tibet, Qinghai, and Kham, showing major salt sources, trade towns, monasteries, mountain passes, and core source-attested corridors.

These exchanges were sustained through a wider geography of caravan routes, markets and social relationships. The concept of *netsang*, or fictive kinship, where each family had trade partners in another specific family in other groups that served as hosts as well, illustrates how barter operated within a moral economy of balance rather than the capitalist motives of profit (Rizvi, 1996). It was adopted between the pastoral and trading communities of *Changpas* and *Brokpas*. The interdependence or movement of salt within a broader commodity system can be studied through the Rupshu & Tso Kar region. The Rupshu are pastoral nomads who subsist on a mixed economy of salt extraction, livestock herding, and the trade of pashm and wool (Ahmed, 1999; Dollfus, 2013).

The interdependence that existed in history forms the foundation of the main hypothesis tested in this paper. Instead of taking the significance of salt for granted due to its frequent mention in history, the question to be asked is whether salt held a connective structural position within the commodity layer of Himalayan commerce.

SALT, EMBEDDED ECONOMIES & FRONTIER GOVERNANCE

Himalayan salt exchange was embedded within social institutions, extending beyond commodity transactions or mere barter systems. The *netsang* system provided security and safety to

Rupshu households in years of poor salt harvest by linking them to established trading partners through hospitality, protection, and reciprocal obligation. Salt supply to monasteries at Korzok and Thugji linked commodity circulation with religious and redistributive institutions (Ahmed, 1999; Rizvi, 1996). These relationships reflect Mauss's discussion of obligation in exchange and Polanyi's argument that economic activity is embedded within society rather than organized through markets alone (Mauss, 1925/2001; Polanyi, 1957).

Tso Kar offers a clear example of Polanyi's substantive economy. Salt extraction followed the annual pastoral cycle of the *Changpas* and Rupshus: autumn was the primary salt harvesting season, summer supported mobility, spring involved birth and migration, and winter restricted productive activity (Ahmed, 1999). Harvesting was organised through male work teams known as *chu lag*, led by *chu dpon*, and tied to the same *pha shun* or patri-fraternal groups, showing the embeddedness of labour within kinship structures (Ahmed, 1999). It was supervised by the *kotwal* who is assistant of *goba* (village head appointed by the king), suggesting that economic and political institutions were intertwined before more formalised regulatory regimes.

The subsistence ethic by Scott provides a different perspective on the pastoral economy by foregrounding risk avoidance, food security, and household survival rather than profit maximisation (Scott, 1976). Scott's theory on pastoral production helps us understand salt production by the Rupshu and *Changpa* tribes. Salt production formed one part of a wider livelihood bundle that included pastoralism, pashm and other wool production, trade caravans, mobility between ecological zones, non-sedentary agriculture, and foraging (Ahmed, 1999; Dollfus, 2013; Rizvi, 1996). Ecological constraints were reflected in this seasonal division: salt production was done during the fall season, grazing and moving took place during summer, while birth and movement took place during the spring, and productive activities were severely limited to winter. More importantly, networks of *netsang* ensured that risk was spread out across the zones, while mutual access to grain ensured a degree of safety in years of low local production (Fisher, 1986; von Furer-Haimendorf, 1975).

In sum, these examples show that salt exists at the juncture between sustenance, reciprocity, and governance. Consequently, political regulation of salt was not only about the flow of a commodity, rather it intervened in pastoral livelihood, custom, resource access, and inter regional trade. This political economic context allows salt to be understood both as a commodity and infrastructure through which Himalayan relations were arranged.

Computational Approaches to Historical Archives

Historical documents typically feature a vast amount of relational data encoded in free text form. Computational techniques like Named Entity Recognition (NER), relation extraction and network reconstruction have been found useful for extracting people, places and organisations from large-scale collections (Ehrmann et al., 2023; Graham et al., 2015; Painter et al., 2019). Various projects, including Six Degrees of Francis Bacon, combine computational inference with human validation, while Beyond 2022 similarly uses structured extraction from archival sources to support historical assessment rather than automatic acceptance of computational findings. (Beyond 2022 Project, n.d.; Warren et al., 2016).

Taking advantage of the above techniques, this study combines automated text processing with manual historical checking. It first identifies relevant entities, then looks for nearby terms and possible relations, validates those relations against the source text, and finally analyses the resulting network. Hence, the contribution of this study does not lie in particular computational tools, but in their combination in answering a certain historical question did salt play a structurally connecting role in Himalayan multi-commodity trade. Consequently, computational networks can be considered as source based representation of relations and not an actual reconstruction of historical reality.

ARCHIVAL TEXT TO NETWORK

Historical sources related to Himalayan trade contain extensive information on commodities, communities, routes and political institutions, but such knowledge is primarily embedded as narrative prose rather than structured representation. The computational approach adopted in this study results in a knowledge graph which translates the evidence spread across the textual sources into a single relational representation. Here, historical entities are represented as nodes while supported relationships are labelled edges. Apart from providing structural relationships within historical texts, the knowledge graph can complement close reading. (Graham et al., 2015; Painter et al., 2019).

This relational approach brings different dimensions of Himalayan salt, such as its provenance, its position within a wider system of exchange, and its circulation with other commodities such as barley, borax, tea etc. into a common analytical structure. The network simultaneously tries to connect salt with pastoral communities, markets, monasteries, and even political treaties. In this respect, the approach also draws lightly upon Actor Network Theory by treating commodities, institutions, communities and political arrangements as relational actors whose significance emerges through their associations with one another (Latour, 2005). More generally, historical network analysis provides a formal means of investigating relationships rather than employing the idea of a ‘network’ only metaphorically a point developed across studies of ancient, Ottoman, European, and imperial archival networks (Alexander & Danowski, 1990; Barkey & Van Rossem, 1997; Błoch et al., 2022; Lemercier, 2012).

The major components of the network representation are nodes organised into broad categories such as commodities, locations, groups, person and concepts. While edges represent specific relations in the form of Subject-Relation-Object, these include economic relations like trade, barter, exchange and supply; political and administrative relations like taxes, governing, regulation. This representation makes it possible to examine associations between human and non-human actors without assuming that historical agency belongs exclusively to individuals or institutions.

The concept of network backbone is used specifically to describe salt's structural position as a commodity in the reconstructed network. Salt is a potential backbone not merely owing to its frequency but its circulation connects other distinct commodities, markets, exchange and creates a whole social ecosystem around its own. From an Actor Network Theory perspective, salt is therefore approached not simply as a passive commodity but as an element whose movement helps organise relations among communities, routes, markets and institutions. Degree, weighted degree and betweenness centrality are used parallelly to assess salt's position, apart from its obvious qualitative composition. This use of formal network properties follows historical network research that distinguishes systematic relational analysis from metaphorical uses of the network concept (Lemercier, 2012; Painter et al., 2019).

Network measures are not considered as direct indicators of historical significance. Instead, they capture the network structure which emerges from the specific corpus and are inevitably influenced by factors such as source survivability, focus on documentary relationships, and relationships which can be discerned from the extant sources. Network history has always known that partial information and lack of full knowledge about past relationships have been key problems for network analysis (Wetherell, 1998). As a result, the network becomes seen as a model derived from sources of past relationships instead of a true picture of the historical economy. Computational measures suggest patterns for historical scrutiny, while the significance of these patterns is verified through the textual data. (Painter et al., 2019).

METHODOLOGY

The study uses a human-in-loop workflow in order to map the relationships among commodities, communities, political institutions as well as geographical locations, based on the texts on Himalayan trade. The methodology has four crucial stages:

(i) Corpus preparation and Data Preparation

(ii) Entity Identification,Human guided Topic selection, co-occurrence and relation extraction

(iii) Graph refinement through configuration updates, entity-type correction and interactive visual inspection

(iv) Corpus-based validation, knowledge graph construction followed by network analysis

Computational extraction identifies relational patterns at scale, while human intervention supports node categorisation, historical relevance, and relation evaluation.

Although applied here to salt, the workflow is domain agnostic and can be adapted to investigate other commodities or focal entities by modifying the corpus, entity categories and relation schema while retaining the same stages of extraction, validation and network analysis.

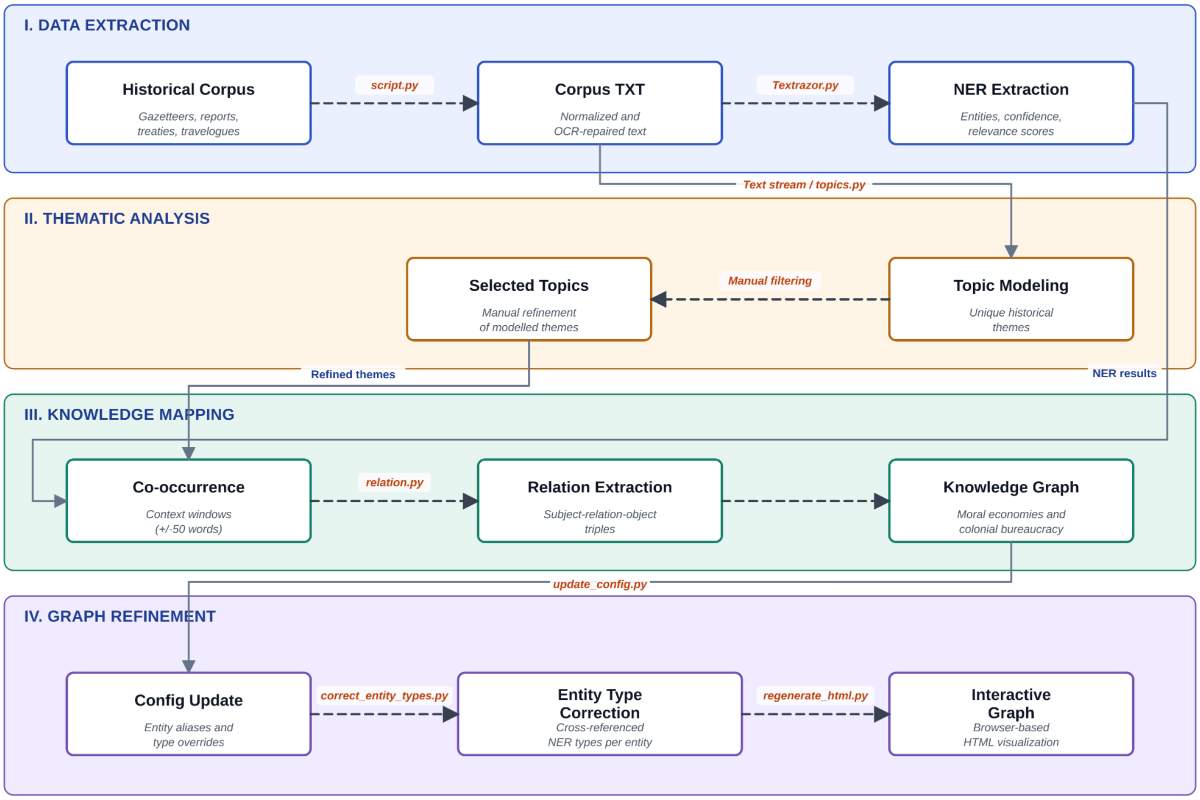

Figure 2. Human-in-the-loop workflow for transforming Himalayan historical sources into a validated knowledge graph, moving from corpus extraction and topic modelling to relation extraction, graph refinement, and browser-based visualization.

CORPUS and DATA PREPARATION

The corpus comprises 20 sources concerning salt, commodity circulation, trade routes, political relations across Western Himalayas and adjoining Tibetan regions. Primary and secondary sources were included to capture contemporary historical accounts alongside ethnographic interpretations. These sources include gazetteers, administration reports, and travel accounts from the nineteenth and twentieth centuries, providing historical context for commercial markets, routes and administrative or political institutions.

Table 1: Bibliography of Historical and Ethnographic Sources

S. No.	Source	Year	Source Type	Region	Classification
1	Western Himalaya and Tibet: A Narrative of a Journey Through the Mountains of Northern India, During the Years 1847-8	1852	Travelogue / digitized book	Western Himalaya, Tibet	Primary
2	Report of a Mission to Sikkim and the Tibetan Frontier; with a Memorandum on Our Relations with Tibet	1885	Administrative memorandum	Tibet frontier	Primary
3	The Himalayan Gazetteer: The Himalayan Districts of the North-Western Provinces of India	1886	Gazetteer	Western Himalayas	Primary
4	Central Asia and Tibet: Towards the Holy City of Lassa, Vol. 1	1903	Travelogue / expedition account	Central Asia, Tibet	Primary
5	Central Asia and Tibet: Towards the Holy City of Lassa, Vol. 2	1903	Travelogue / expedition account	Central Asia, Tibet	Primary
6	Report on Tibet	1903	Administrative report	Tibet	Primary
7	Gazetteer of the Mandi State	1908	State gazetteer	Mandi	Primary
8	Imperial Gazetteer of India: Provincial Series, Kashmir and Jammu	1909	Imperial gazetteer	Kashmir, Jammu	Primary
9	British Garhwal: A Gazetteer	1910	District gazetteer	Garhwal	Primary
10	The Salt Industry in India	1956	Administrative / industry report	India	Primary
11	Himalayan Traders: Life in Highland Nepal	1975	Ethnography / monograph	Highland Nepal	Secondary

S. No.	Source	Year	Source Type	Region	Classification
12	Adaptation to a Changing Salt Trade: The View from Humla	1982-83	Journal article	Humla, Nepal	Primary
13	Trans-Himalayan Traders: Economy, Society, and Culture in Northwest Nepal	1986	Ethnographic source	Himalayan trade regions	Secondary
14	Ladakh: Crossroads of High Asia	1996	Regional historical source	Ladakh	Primary
15	The Salt Trade: Rupshu's Annual Trek to Tso Kar	1999	Book chapter	Rupshu, Ladakh	Secondary
16	Becoming India: Western Himalayas under British Rule	2007	Historical monograph	Western Himalayas	Secondary
17	No longer tracking greenery in high altitudes: Pastoral practices of Rupshu nomads and their implications for biodiversity conservation	2013	Journal Article	Ladakh	Secondary
18	Transformation Processes in Nomadic Pastoralism	2013	Journal article	Ladakh	Secondary
19	Salt Routes and Barter Caravans in the Himalayan Regions of Nepal and Tibet from an Ethnographical Perspective	2022	Book chapter	Tibet, Nepal Himalaya	Secondary
20	The Salt Trips in Tibet and the Himalayas: Extraction and Trade in Pre-modern Times	2022	Journal article	Tibet, Himalayas	Secondary

The mixed corpus frames salt as a relation-making object linking trade, mobility, taxation, dispute and customary exchange.

Different tools were used to prepare these historical sources into usable data. These sources were first converted into machine-readable text using PyMuPDF, with Tesseract OCR fallback for image-based pages (i.e. pages with insufficient extractable text). Preprocessing was kept minimal to preserve historical names and regional terminology.

After experimenting with various NER models, the approach uses the TextRazor API to improve named entity recognition and topic detection in the corpus. Entities extracted are categorized in terms of their relevance and confidence ratings, while high level topics of the documents are also extracted. Confidence score reflects the model's degree of surety that the particular entity was correctly recognized, especially when dealing with rare or domain-specific entities like *Changpa* (people) and Zanskar (geolocation).

Relevance of the entity reflects its degree of importance in relation to the topics of the document, and allows for filtering out commodities, trading organizations, tribes, and other relevant structures in order to determine whether salt routes provided the necessary infrastructure for multi-commodity trade. In combination with confidence ratings, these two values allow differentiating between trade-related concepts of core interest and peripheral mentions: entities with high relevance and confidence form the core concepts for analysis, while high-confidence and low-relevance entities form reliable but peripheral observations, and vice versa.

ENTITY IDENTIFICATION & RELATION EXTRACTION

Automated topic identification through Named Entity Recognition produced 957 cumulative semantic categories, many of them being unrelated to the area of study. Therefore, a human-in-loop filtering stage reduced these to 57 research-selected categories covering dominant topics like geographical features, commodities, jurisdictions, organisations, transport, trade, economy etc. These topics provide interpretive anchors for understanding how salt is embedded with large material and political circuits.

Moving beyond the entity list and topics, the paper aims to identify relations in the form of (Subject, Relation, Object) structure within text. Examples of such relational triples include:

'Rupshu nomads — extract salt from — Tso Kar'

"Mandi State — imposes tax on — salt caravans"

"Ladakh royal trader — permitted entry to — Rudok (per Treaty of Tingmosgang)"

These relations are grounded in a word co-occurrence window approach, which uses anchor entities such as salt, wool, Rupshu, or Changthang to identify the strength and frequency of surrounding terms and their conceptual proximity. For each occurrence, the workflow used a symmetric window of approximately 50 words on either side and recorded other recognised entities occurring. This method helps identify thematic clustering around each term.

Potential textual segments were then analyzed utilizing a local deployment of the Llama **3** model via the **Ollama** framework. Relation extraction was confined to a specific set of semantic categories characterizing political-administrative, structural, and economic connections, such as *trades_with*, *exchanges_for*, *extracts_from*, *transports_via*, *taxes*, *controls*, *governs*, *licenses*, *supplies*, *depends_on*, and *migrates_through*. The resulting relational patterns were structured as directed **Subject–Relation–Object triples**, preserving the original source text for subsequent verification and quality control. Duplicate triples were consolidated, with their frequency recorded as an edge weight for network analysis.

GRAPH REFINEMENT

Before validation and analysis, however, the extracted graph was improved, since there were spelling differences, duplicate names, label inconsistency, and classification issues that had occurred during the automated extraction of entities. This process was particularly important as historical entities tend to occur in variant spellings and classifications in various types of sources like gazetteers and travelogues.

Improvement of the graph included modification of the configuration file with aliases and types of entities and reclassification of some entities into more appropriate classes followed by re-creation of the interactive graph. Web visualization allowed for the detection of isolated instances, duplicates, impossible relationships, and category conflicts. Graph improvement did not substitute source validation but only prepared the extracted graph for further analysis.

VALIDATION & NETWORK ANALYSIS

The final product from 18 sources turns out to be 818 relation rows, of which 654 were classified as Validated, 98 as Probable and 67 as Missing / Unsupported. Validated relations have direct textual support, probable relations are plausible but contain contextual and relational ambiguity, whereas missing relations lack sufficient evidence. These categories function as quality control procedures rather than as direct measures for model accuracy.

Validated triples were combined into a network in which entities are nodes and labelled relations are edges. Entities were normalized and categorized into groups such as locations, goods, groups, persons and concepts, whereas repeated relations were retained as weighted edges.

The network was analyzed through degree, weighted degree, betweenness centrality, and the Louvain community detection method. Degree refers to the number of connections that an entity has, weighted degree takes into account the frequency of connections, and betweenness centrality reflects how an entity serves as a bridge (Freeman, 1977; Newman, 2010). The Louvain method uses modularity optimization to detect communities where there is more connectivity within the group than with entities outside the group (Blondel et al., 2008). Since the output from this method could be different each time, it was run several times, with modularity values and number of communities compared.

The network statistics are understood as characteristics of the corpus-derived relationship network, not as indicators of historical significance. Therefore, quantitative analysis should always be considered in conjunction with the primary data.

RESULTS

EXTRACTION & NETWORK CHARACTERISTICS

The final corpus is composed of 20 sources including colonial gazetteers, travel and administrative accounts concerning Himalayan trade, especially salt circulation. Relation extraction generated 1,475 edges, of which 1,227 were validated against source documents, while 138 were classified as probable and 110 as missing or unsupported. The aggregated graph had a total of 1171 nodes and 973 distinct edges with 141 connected components. The largest component consists of 836 nodes with network density of 0.0188 and average degree 1.73.

These results suggest the graph to be sparse but structurally differentiated, which is expected in historical relation networks derived from documents. Not every entity is supposed to be connected to another entity. The connectedness of the majority of nodes into a single large component is indicative of capturing a broad relational field rather than collection of entirely isolated cases.

Table 2. Network level characteristics

Graph View	Nodes	Edges	Components	LCC Size	Density	Avg. Degree	Louvain Communities
Validated	999	958	135	692	0.0019	1.92	22 communities; Q = 0.815
Historical Entity	509	422	112	229	0.0033	1.66	13 communities; Q = 0.762
Trade-oriented	637	591	103	384	0.0029	1.86	17 communities; Q = 0.786

The validated network is sparse but not scattered showing that the corpus preserves a broad relational field around Himalayan trade and governance.

Modularity (Q) is defined by how much a graph is subdivided into densely connected communities with little connectivity among them. **Table 2.** shows strong community structure across all three graph views: the validated view contains 22 communities ((Q = 0.815)), the historical entity view 13 ((Q = 0.762)), and the trade-oriented view 17 ((Q = 0.786)). Louvain community detection returned a median number of 24.5 communities and a median modularity of 0.819, suggesting that there is community structure where the nodes have been clustered in distinct relational clusters and not into one homogeneously connected network (Blondel et al., 2008). This aids in identifying the brokers between communities, whereas the disconnected components remind us that this is a representation of recoverable relationships and not an entire reconstruction of the Himalayan political economy (Painter et al., 2019; Wetherell, 1998).

MULTI COMMODITY NETWORK

Salt occupies the most prominent position among commodity labelled nodes. The validated graph suggests salt as a node to have a degree of 44, a weighted degree of 87 alongside 0.2173 and 0.0135 betweenness, centrality and page rank respectively. Degree and strength measure indicates salt's breadth and recurrence of connection, while betweenness indicates its brokerage role within the network.

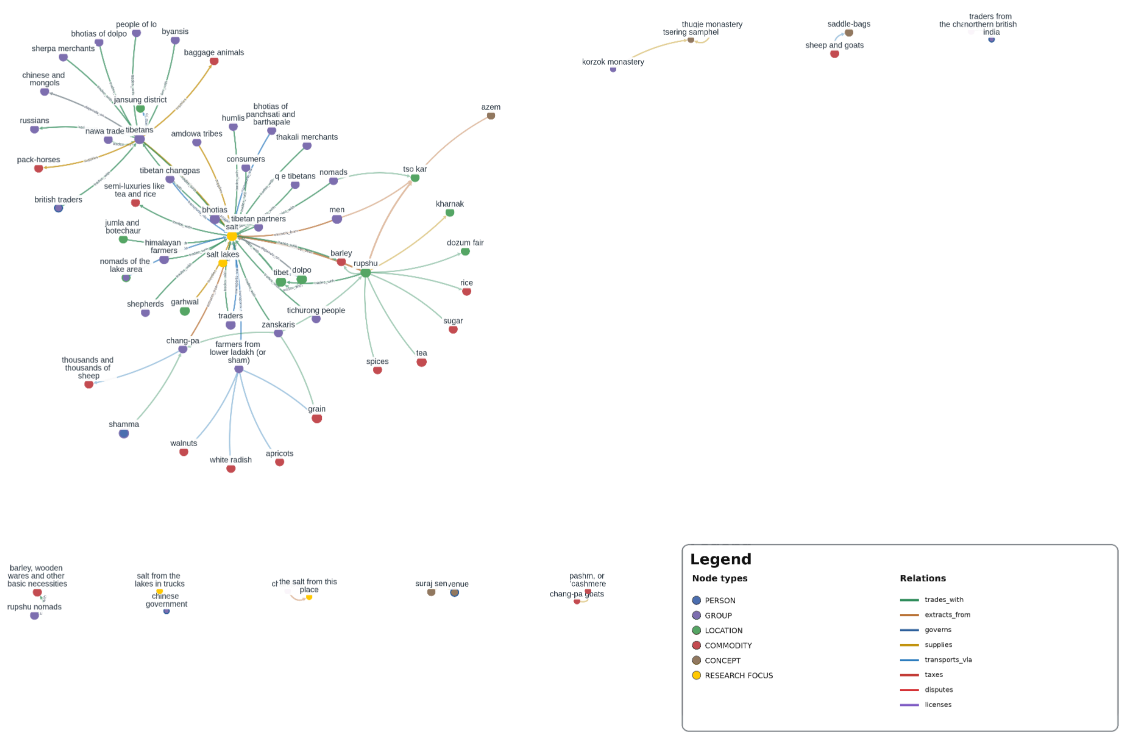

Figure 3. Salt-centred commodity network showing salt's relational links with trade goods, communities, places, routes, and political actors across the Himalayan frontier.

Commodity label permutation tests distinguish salt from other commodities. Across 100 permutations, salt's observed degree, page rank, strength and betweenness and removal effect exceeded the corresponding null distribution with p value = 0.0099 ($p < 0.05$). These results show salt to be the most prominent commodity node in the whole graph, thus a *keystone commodity* of the region’s political economy.

Table 3. Centrality scores for major commodity nodes, comparing degree, weighted strength, betweenness, and PageRank

Commodity	Degree	Strength	Betweenness	PageRank
Salt	44	52	0.2173	0.0135
Yak	13	14.0	0.090	0.0047
Grain	12	12.0	0.071	0.0041
Tea	7	7.0	0.025	0.0028
Wool	5	5.0	0.027	0.0017
Pashm	4	5.0	0.012	0.0017
Barley	4	4.0	0.007	0.0014

Salt stands out because its centrality is not only high in frequency or degree, but also in brokerage. Its ties across commodities, communities, routes and institutions make it the strongest structural candidate for the paper's claim that salt functioned as a *keystone commodity* and a political infrastructure.

Node removal analysis adds another dimension to this interpretation. Removing the salt node breaks the single large component into 18 additional parts (132 to 150 in the whole graph) and lowers global network efficiency. This does not imply that the network collapses without salt, rather indicates salt to be a broker entity binding the network better than any other; guarding against fragmentation and low connectivity.

Salt’s participation coefficient of approximately 0.468 indicates its distribution across relational communities and its capacity to link commodities, regions, and social clusters. Edge-level connections with Tibetan and Bhotia traders, Thakalis, Tarangpurians, and others suggest that this centrality spans several geographically and socially distinct exchange circuits. Taken together, these findings identify salt as the most structurally central commodity in the validated knowledge graph and, therefore, as a *keystone commodity* in the region’s political economy.

GOVERNANCE & POLITICAL EXCHANGE

The network contains political and economic relationship labels showing how commodity circulation was regulated and organised. Labels such as governs, taxes, controls, licenses, and monopolizes relate to rule making, resource collection, control of access, regulated movement/production, and trade privilege. The terms depends_on, supplies, transports_via, and trades_with pertain to the economic and logistical connections that allowed for this authority to be established. As an example, Zhao Erfeng governed and collected taxes from the Tsakhalho region regarding the transportation and taxation of salt, and the Nepal government imposed taxes on goods such as salt, wool, textiles, and goats.

These computational connections reflect a longer recorded history in which there were political connections to commodity flow prior to colonialism. The Treaty of Tingmosgang of 1684 established salt and related trade privileges as an instrument of political control long before the colonial interventions (Treaty of Tingmosgang, 1684). It granted exclusive rights over the *Ngarees Khorsum* wool trade to Ladakhi merchants and restricted access to Rudok to the king's royal traders, showing that political regulation of trade preceded British intervention.

Nonetheless, the later archive indicates growing bureaucracy concerning the administration of salt through taxation, licensing, monopoly and supervision. The early versions of the project were also able to differentiate between reciprocity and customarily-based circulation on one hand and the more bureaucratic form of circulation which became more common in later nineteenth and twentieth-century records. These findings demonstrate that the political sphere of the knowledge graph is intertwined with the commodity circulation directly. The consequences of such transition are particularly interesting in terms of the relationship between.

RUPSHU–TSO KAR (CASE STUDY)

The Rupshu-Tso Kar area offers an opportunity to investigate how the computational network aligns with historically documented exchange relations. The primary sources indentify Tso Kar as an important salt-production site embedded in pastoral mobility and multi-commodity exchange, where salt, wool, and pashm were traded for barley, grain, and other necessities (Ahmed, 1999; Dollfus, 2013; Rizvi, 1996, 1995, 1999; Thomson, 1852).

The network includes some aspects of this configuration as salt is linked to nomads, lake sites and exchange in the region. Nonetheless, the degree of support varies across relations within the configuration. While some extraction relations have retained evidence, several Rupshu-specific edges such as the dispute between Kharnak and Tso Kar lack it.

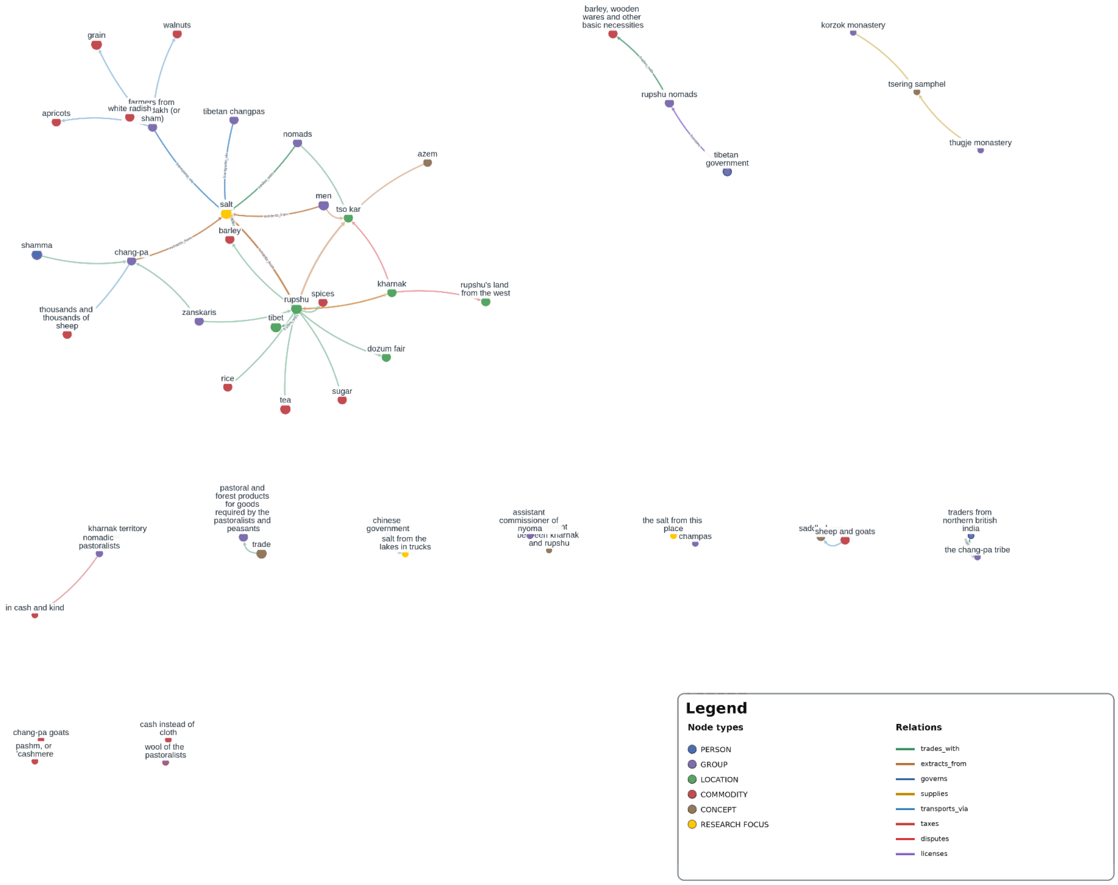

Figure 4. Rupshu-Tso Kar case-study network highlighting salt extraction, pastoral mobility, regional exchange, and disputed relations around the Tso Kar salt landscape.

At the same time, the historical sources independently describe the process of seasonal salt extraction at Tso Kar, local control through *goba* and *kotwal*, exchange with adjacent areas and disputes over the lake (Ahmed, 1999; Dollfus, 2013). The case therefore shows both the value and the limits of the computational model: it can reconstruct clusters of extraction and exchange, but individual edges still require source-based verification (Painter et al., 2019; Wetherell, 1998).

DISCUSSION

SALT AS NETWORK BACKBONE

Though the results support salt as a structural component of Himalayan commodity exchange, the network doesn't demonstrate it to be dominating every form of commerce across Himalayas. Rather, salt emerged as a structurally central commodity with a relatively high degree and weighted degree for breadth and recurrence of its connections. Several permutation tests, betweenness scores, and salt's participation across communities suggest that it functions as a broker entity in the restricted graph (Blondel et al., 2008; Freeman, 1977; Newman, 2010). Removing the salt node also suggests the same owing to the increase in the number of connected components of the graph.

Historical evidence provides an additional layer of substance to the computational pattern. The Gazetteer of Kangra District documents the flow of Tibetan salt to Lahul, which was exchanged for barley and taken further to Kulu, and the same routes were responsible for Tibetan wool, apricot stones, mustard-seed oil and other products (Government of Punjab, 1917). There are similar relations in the computational graph, such as the connection between Bhotia traders and Tibetan salt with grain-growing regions of Upper Garhwal (Walton, 1910).

Salt's centrality can also be understood via ecological complementarity. High-altitude salt producers exchanged salt for agricultural valley goods such as barley and grain. The Tso Kar region along with the salt, livestock, wool, and pashm attracted caravans from neighbouring agricultural and trading regions (Ahmed, 1999; Lecoq, 2022; Rizvi, 1999). These movements formed overlapping commodity circuits, where salt was exchanged repeatedly against goods, among communities, and at different locations (Fisher, 1986; von Furer-Haimendorf, 1975).

COMMODITY TO POLITICAL INFRASTRUCTURE

Salt's relations with taxation, control, and administration place it within the network of political authority. The Treaty of Tingmosgang is a clear indicator that in the 17th century, commodity exchange between Ladakh and Tibet was already subject to institutionalized governance (Rizvi, 1996; Treaty of Tingmosgang, 1684). In particular, the *Ngarees-khor-sum* wool trade was limited to Ladakhis, entry into Rudok was limited to the royal traders of Ladakh and the Lhasa tea and *Lo-chhak* mission from Leh were tied to Ladakhi obligations. Therefore, salt established access to markets, territorial movement, commodity privileges, and politics-religious authority.

Salt also entered increasingly explicit administrative forms. For example, the salt mines of Guma and Drang constituted an important source of revenue for the Mandi state. By the 19th century, Mandi's administration involved a specialised Salt Department, inspectors, mine officials, weighing personnel, and attendance registers. The 1904 Gazetteers records fixed prices, duties, revenue sharing contracts between Raja and the British government, and British sanctions (Emerson, 1904; Foreign Department, Political Branch-A, 1874).

Colonial rule did not introduce politics into an apolitical authority, rather altered the institutional form of operation. Earlier arrangements had been organised around royal privileges, customary authority, patronage, monasteries, reciprocal obligations etc (Ahmed, 1999; Rizvi, 1996). Colonial administration involved standardised, bureaucratically documented contracts (Alam, 2007; Emerson, 1904). Salt functioned as a political infrastructure, because control over it meant access to resources, movement, labour, taxation, and territorial authority. This also aligns with Actor Network Theory, since the political implications of salt trade emerged through relations among lakes, products, pastoralists, animals, monasteries, officials, weights, and documents, rather than through either state or market alone (Latour, 2005).

From embedded exchange to colonial legibility

Polanyi’s concept of embeddedness is useful here because the sources do not show a clear division between market and non-market activities. Rather, economic activity was embedded in social, ecological, and political institutions (Polanyi, 1957). Thus, at Tso Kar, extraction was organized through seasons, work teams such as *chu lag*, *chu dpon*, *kotwal* and *goba*, and *netsang* helped maintain relationships based on hospitality, protection and trade relations (Ahmed, 1999; Rizvi, 1996). Production and circulation were therefore embedded in kinship, custom, and mobility which is the view taken by Polanyi's substantivist theory.

Gazetteer evidence also shows the coexistence of embedded and monetary exchanges. In Lahul, grains and salt could be exchanged at fixed ratio, while Gaddi shepherds provided sheep to villagers in return for storing grain and salt for future visits (Government of Punjab, 1917). These exchanges combined calculation with recurrent social obligations. They should not be idealized as transactions devoid of calculation, but instead illustrate that calculation occurred within long-term structures of mobility, reciprocity, and interdependence (Mauss, 1925/2001; Polanyi, 1957).

The Mandi sources present another logic of visibility (Emerson, 1904). Salt gradually turned out to be a commodity which was to be priced, measured, assigned a destination, divided into duties between governments, and controlled by specific officials. The gazetteer describes functionally distinct officials responsible for mining, attendance, tools, and measurement, along with formal arrangements of protection (Emerson, 1904). This supports Scott’s idea of legibility: complex local practices became more governable when represented through standard units of classification, measurement, record-keeping, and administrative flow (Scott, 1999).

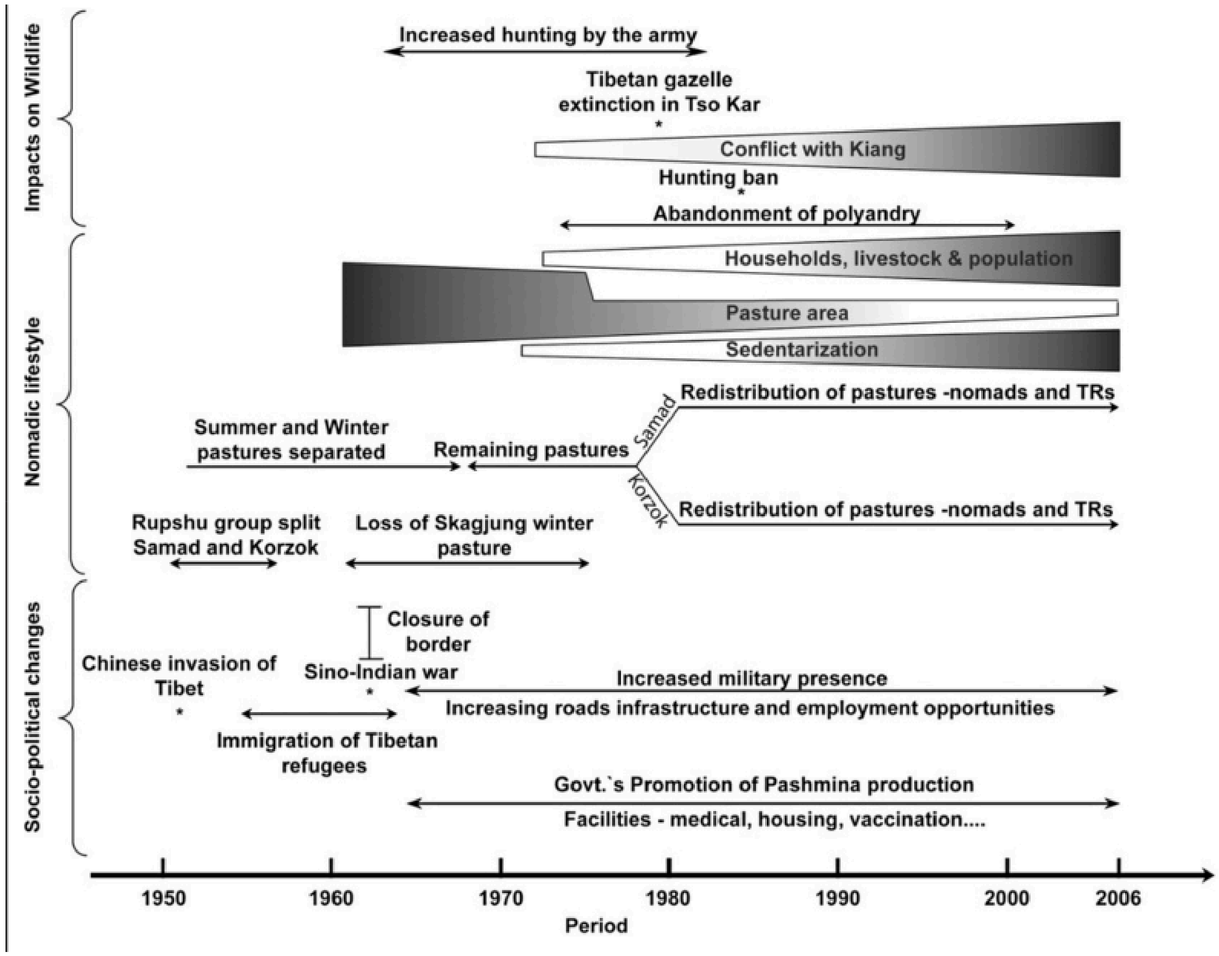

Figure 5. Chronology of socio-political, pastoral, and wildlife-related changes in the Changthang region from the 1950s onward. Adapted from Singh, N. J., Bhatnagar, Y. V., Lecomte, N., Fox, J. L., & Yoccoz, N. G. (2013). ***No longer tracking greenery in high altitudes: Pastoral practices of Rupshu nomads and their implications for biodiversity conservation*****.** ***Pastoralism: Research, Policy and Practice, 3*****, Article 16. https://doi.org/10.1186/2041-7136-3-16**

This transformation was uneven rather than a total replacement of one economic system by another. The practice of barter, pastoral mobility, and reciprocity persisted in sources from periods of increasing bureaucracy (Government of Punjab, 1917; Walton, 1910). This would then imply a shift from primarily embedded and mediated exchange toward more formal and bureaucratized regulation, with both systems sometimes existing side by side. Additionally, Thompson's differentiation of material standards and a 'way of life' – improvements in bureaucracy and commercialization do not necessarily indicate improvement or deterioration in local welfare (Thompson, 1963).

In light of these findings, salt's importance lies not only in its prevalence but in its ability to link ecological zones, commodities, communities, institutions and mechanisms of control and access. The network does not make salt the backbone of the economy of the Himalayas. Instead, it provides support for the fact that salt is the most structurally central commodity in the reconstructed corpus and, therefore, a unique way through which we can understand the evolving dynamics of embedded exchange and frontier rule.

Beyond salt, this domain-agnostic workflow can be adapted to other commodities or focal entities, while future work should test its scalability on a broader, more systematic corpus with greater primary-source coverage across regions and periods.

LIMITATIONS

The reconstructed network is conditioned on the composition of historical corpuses such as gazetteers, administrative reports, travel accounts etc. These sources were produced for different purposes and provide uneven coverage of communities, regions, and commodities. Therefore, the resulting graph is not an exhaustive representation of Himalayan trade. Any absent relationship may be directly pointing to the gaps in the corpus.

Computational processing also introduces a second set of limitations. TextRazor's underlying knowledge base may not detect historical obscure entities such as person or place. Though the human guided filtering of relevant topics improves precision, it adds an explicit interpretative layer into the extraction process. Similarly, the ±50 word co-occurrence or textual proximity is limited to recognize relationships spanned across longer passages. Relations extracted by Llama 3 are also susceptible to being misidentified. Of 1475 extracted relations subjected to validation, 1227 (83.2%) were classified as Validated, 138 (9.4%) as Probable, and 110 (7.5%) as Missing/Unsupported.

Lastly, network metrics are a characterisation of the structural nature of the computational representation of the source material and not the historical economic system independent of documentation thereof. Central or bridging locations can thus be indicative of both the historical connections and the documentation thereof. Network studies in history thus entail that nodes, edges, and weights are to be interpreted within the context of the production process as well as the computational process of the sources. Network analysis is thus seen as complementary to source criticism and historical interpretation and not as a replacement thereof. Assertions regarding the structural location of salt are thus confined only to those connections recoverable from the analysed corpus.

CONCLUSION

Through this paper, it has been demonstrated that rather than being only a commodity, Himalayan salt was an agent for connecting people politically and economically in the Western Himalayas. Through the combination of historical sources such as the archival records, colonial gazetteers, treaties, and computational text analysis, it has been shown that the salt routes connected pastoral mobility, regional exchange, state power and multi-commodity trade (Ahmed, 1999; Rizvi, 1996; Foreign Department, Political Branch-A, 1874; Marshall, 2004).

The historical process of transition from netsang-based reciprocity to colonial contracting and regulatory regime illustrates the transformation of Himalayan frontier into one more knowable and governable space (Bergmann, 2016). The regulation of salt production, taxation, and mobility by the British changed the existing frontier relations and resulted in the emergence of new forms of sovereignty in the margins of empire (Aggarwal, 1956; Syed, 2010). However, the process has been far from being uniform as it involved both pastoral adaptation and the struggle for the lake resources in the larger context of Tibetan, Ladakhi and British politics.

This is supported computationally by the importance of salt, its brokerage function, and participation in the relational communities. Relation Extraction, Topic Modelling, and NER reveal the layers of Himalayan trade relations that include states, pastoralists, ecology, and commodity chains. Hence, the paper provides a computational way based on sources for the analysis of salt trails as political infrastructures and how substances such as salt contributed to the territory, society, and economy of the Himalayas (Fisher, 1986; Lecoq, 2022).

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